



EESTech Challenge LC Aachen

Eric Schulze

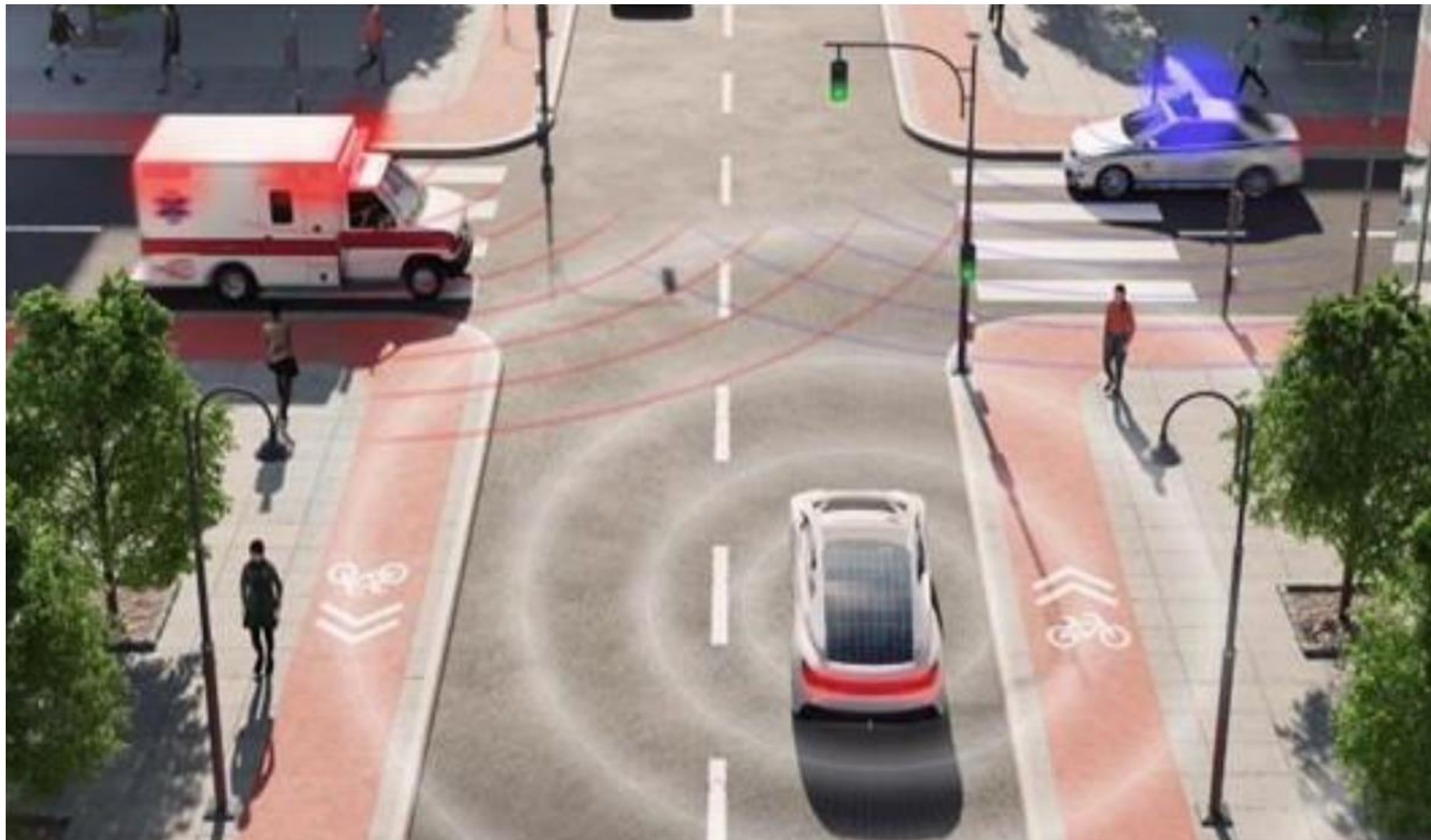


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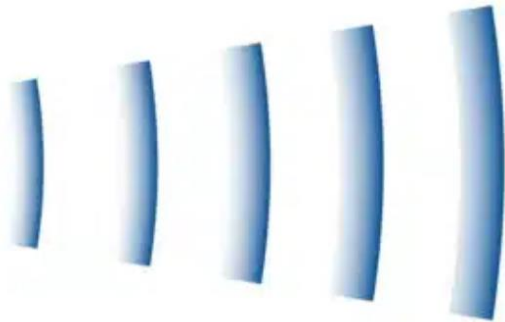
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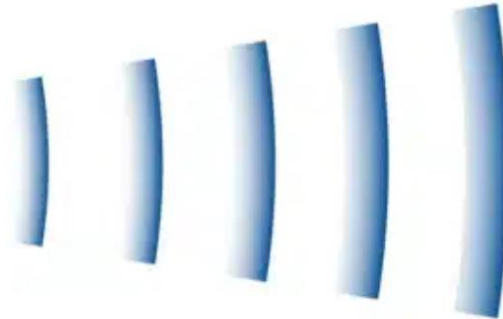
Introduction Challenge

- Motto: Autonomous Driving and Secure Connected Vehicles
- In this challenge, participants are tasked with developing an AI-based solution to detect the direction of an approaching emergency vehicle.
- Using the PSOC™ Edge AI Kit, your goal is to build a system that can recognize from which direction a siren or emergency signal is coming. The more accurately and reliably different directions are detected, the better your solution will perform.

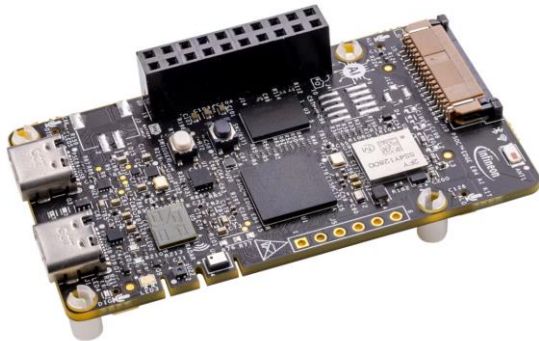
Introduction Challenge



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- Develop an AI model that determines the direction of an emergency vehicle which runs on the PSOC edge
- Achieve high accuracy across multiple directional inputs.
- Test and Flash it on embedded hardware (PSOC™ Edge E84 AI Evaluation Kit)

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Criteria	Comment	Weight
Task 1	- Task 1 - Develop an AI model that determines the direction of an emergency vehicle which runs on the PSOC edge AI Kit - Goal is to have many as possible direction and to be accurate as possible. $\text{Score} = \left(\frac{1}{n} \sum_{i=1}^n \frac{r_i}{5} \right) \cdot (1 - e^{-0.5n})$	50%
Presentation	Present your results. 😊 3-5 min	20%
Innovation	Be creative, try to make this challenge easy as possible for you. This can be anything For example automate the process	15 %
Documentation	Write a short documentation of what you did and why. document also the innovative part there. 😊 Send to us the documentation and code via E-Mail or create your own github repo and send us the link	10%
Feedback	Please provide feedback to us. You can write the feedback on github or send it to us via e-mail (ericjoerg.schulze@infineon.com)	5%

$$\text{Score} = \left(\frac{1}{n} \sum_{i=1}^n \frac{r_i}{5} \right) \cdot \left(1 - e^{-0.5n} \right)$$

- n = how many directions
- Ri = how many of 5 tries are correct (for each direction are 5 tests)

Let's Get Started!



github.com/Infineon/hackathon

